ND280 Magnetic Field Monitoring System

Technical Design Report

Version 0.6

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August 2026

Internal development document

Document Control

Document	ND280 Magnetic Field Monitoring System – Technical Design Report
Version	0.6
Status	Draft for internal review
Author	Stefano Levorato
Institution	INFN – Laboratori Nazionali di Legnaro
Repository	https://github.com/stefanolevorato/ND280-Magnetic-Field-Monitoring-System

Revision History

Version	Date	Author	Description
0.1	August 2026	S. Levorato	Initial LaTeX project structure
0.2	August 2026	S. Levorato	Scientific motivation and bibliography
0.3	August 2026	S. Levorato	System Views and Chapters 4–5 completed
0.4	August 2026	S. Levorato	Added monitor long-duration evidence and Rev.A hardware render
0.5	August 2026	S. Levorato	Added calibration and metrological characterization strategy
0.6	August 2026	S. Levorato	Added preliminary stability characterization and ROOT analysis plots

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Chapter 1

Introduction

1.1 Background and Motivation

The ND280 Magnetic Field Monitoring System is an instrumentation project developed by the Istituto Nazionale di Fisica Nucleare (INFN) with the objective of designing, implementing and validating a distributed magnetic-field monitoring infrastructure for the ND280 detector at the Japan Proton Accelerator Research Complex (J-PARC).

The project originates from the need for a compact, modular and scalable monitoring system capable of continuously monitoring the magnetic field at multiple locations inside the detector while providing long-term stability, comprehensive diagnostic capabilities and straightforward integration with the experiment control infrastructure.

Commercial magnetic-field monitoring systems generally do not satisfy the geometrical, mechanical and integration constraints required for the ND280 Near Detector of the T2K experiment, installed inside the C-shaped UA1 magnet. For this reason, a dedicated monitoring system has been specifically designed and developed.

The proposed system is based on intelligent acquisition nodes, each capable of locally acquiring, processing, validating and transmitting magnetic-field measurements while operating autonomously. Each node performs local sensor management, data acquisition, calibration, diagnostics and health monitoring before making the acquired information available to higher-level controllers.

1.2 Design Philosophy

The overall architecture has been intentionally developed following a modular engineering approach in which hardware, embedded firmware, desktop software and technical documentation evolve together under version control.

The system has been designed by separating hardware abstraction, device drivers, acquisition services and application logic. This modular organization minimizes software dependencies and allows future hardware revisions to reuse the same firmware architecture with minimal modifications.

The desktop application, referred to throughout this document as the *ND280 Monitor*, has been developed as an engineering tool supporting commissioning, diagnostics, calibration, visualization and long-term data logging.

The development follows an incremental engineering methodology in which every hardware revision is accompanied by the corresponding firmware, desktop software and validation campaign before introducing additional functionality.

This approach minimizes technical risk, simplifies debugging activities and preserves software compatibility throughout the evolution of the project.

1.3 Project Evolution

The development started from a Reference Single-Sensor Node whose purpose was the validation of the complete acquisition chain, including hardware, firmware, communication protocol and monitoring software.

Once the architecture had been validated through extensive functional and long-duration tests, the project evolved towards the design of an Eight-Sensor Multiplexing Node based on the Texas Instruments TCA9548A I^2C multiplexer.

The new hardware architecture extends the acquisition capability from one to eight magnetic sensors while preserving the existing NodeOS architectural principles.

The long-term objective of the project is the realization of a distributed magnetic-field monitoring infrastructure composed of multiple intelligent acquisition nodes interconnected through a segmented communication network and coordinated by a dedicated controller.

1.4 Scope of this Document

This Technical Design Report describes the complete design of the ND280 Magnetic Field Monitoring System, from the validated Reference Single-Sensor Node to the current Eight-Sensor Multiplexing Node and the future distributed acquisition architecture.

The document presents the hardware design, embedded firmware architecture (NodeOS), communication protocol, desktop monitoring software, validation campaign, engineering decisions and future development roadmap.

The purpose of this document is not only to describe the current implementation but also to record the engineering rationale behind the adopted architectural choices, providing a technical reference for future hardware revisions and software developments.

Chapter 2

Scientific Motivation

2.1 Experimental Context

The T2K experiment is a long-baseline neutrino-oscillation experiment located at J-PARC, Japan. Its off-axis near detector, ND280, is installed approximately 280 m from the neutrino production target and operates inside a magnet providing a nominal field of approximately 0.2 T. The neutrino beam is first measured by the near detectors, including ND280, and then travels to the far detector, Super-Kamiokande.

The two new High-Angle Time Projection Chambers (HA-TPCs) were installed as part of the ND280 upgrade, which was completed at J-PARC in 2024 [1].

ND280 is installed inside the UA1/NOMAD magnet, which provides a nominal magnetic field of approximately 0.2 T. The magnetic field allows ND280 to bend charged-particle trajectories, determine the sign of the particle charge, and measure the momentum of particles produced in neutrino interactions [3, 4, 5].

2.2 Magnetic Field

A TPC does not measure the momentum of a charged particle directly. It measures a sequence of points along the particle trajectory. In a magnetic field, the trajectory is curved, and the curvature is used to infer the momentum and the sign of the charge.

For a charged particle in a uniform magnetic field, the transverse momentum is approximately related to the radius of curvature by

$$p_T\,[\mathrm{GeV/c}] \simeq 0.3\,|q|\,B\,[\mathrm{T}]\,R\,[\mathrm{m}], \tag{2.1}$$

where p_T is the transverse momentum, q is the charge in units of the elementary charge, B is the magnetic-field strength, and R is the radius of curvature.

In the real reconstruction, the track fit should use the three-dimensional field map

$$\vec{B}(x, y, z), \tag{2.2}$$

rather than only the nominal central value. A dedicated device was developed to measure and map the magnetic field of the ND280 magnet [2].

2.3 Magnetic Field at ND280

The value of 0.2 T is a nominal or average value. The actual magnetic field can vary spatially and can include components transverse to the nominal field direction. Variations may occur near magnet openings, detector structures, and the boundaries of the active TPC volume.

For track reconstruction, the relevant quantity is related to the field integrated along the trajectory. Schematically,

$$\Delta\phi \propto \int B_\perp \, \mathrm{d}l, \tag{2.3}$$

where $B_\perp$ is the component responsible for the measured bending and the integral is taken along the track.

A recent HA-TPC performance paper mentions a magnetic-field map obtained from measurements performed in 2009 in the region of the vertical TPCs [1]. This makes it important to verify that the map used for the HA-TPC reconstruction is representative of the actual position and configuration of the upgraded chambers.

2.4 Impact on Momentum Reconstruction

If the field is treated as uniform and the relative uncertainty on its scale is $\delta B/B$, then, for a fixed measured curvature,

$$\frac{\delta p_T}{p_T} \simeq \frac{\delta B}{B}. \tag{2.4}$$

Thus, a one-percent systematic error in the magnetic-field scale produces approximately a one-percent systematic error in the momentum scale. This is a momentum-scale uncertainty and is distinct from the statistical momentum resolution, which is also affected by the number and precision of measured points, diffusion, and multiple scattering.

An inaccurate field map can produce several effects:

- a bias in the reconstructed momentum scale;
- a position-dependent or angle-dependent momentum bias;
- a degradation of the momentum resolution;
- differences between the reconstructed response of positive and negative tracks;
- a lower reliability of the charge-sign determination.

These effects can propagate to the reconstructed lepton momentum and angle, neutrino-energy estimators, event classification, and the systematic uncertainties used in the T2K oscillation analysis. ND280 is used to constrain uncertainties in the neutrino flux and neutrino-interaction models before oscillation [1].

2.5 Specific Relevance for the HA-TPCs

The HA-TPCs were designed to reconstruct particles emitted at large angles relative to the neutrino-beam direction. They are positioned above and below the Super-FGD. The coordinate system used for ND280 is defined with the x axis parallel to the main component of the magnetic field, the y axis opposite to the direction of gravity, and the z axis along the incoming neutrino beam [1].

The magnetic-field map is particularly important for the HA-TPCs for the following reasons:

1. Tracks can cross different regions of the HA-TPC, so the reconstruction depends on the field along the full track.
2. Components of the field that are not aligned with the nominal direction can modify the three-dimensional curvature.
3. Some high-angle particles may have a shorter track lever arm, making the momentum more sensitive to field-scale and alignment errors.

4. The magnetic field can affect electron transport and transverse charge diffusion in the TPC. These effects must be separated from distortions caused by the electric field, field-cage geometry, gas properties, and Micromegas response.

5. The field map enters both the detector simulation and the track-reconstruction algorithms.

2.6 Recommended Checks for an HA-TPC Field Study

A dedicated study should verify:

1. the three-dimensional field map used by the reconstruction software;
2. whether the map is based on direct measurements, magnetostatic simulations, or both;
3. coverage of the complete HA-TPC active volume;
4. the absolute accuracy of the field scale;
5. the accuracy of the B_x, B_y, and B_z components;
6. the alignment between the field-map coordinate system, the magnet, and the HA-TPCs;
7. agreement between cosmic-ray data and simulation;
8. comparisons of positive and negative tracks;
9. reconstructed momentum as a function of position, angle, and charge;
10. propagation of the field uncertainty into the final physics analysis.

2.7 Implication for the Monitoring System

Measuring the magnetic field in ND280 is necessary because the momentum and charge sign are obtained from the curvature of charged-particle tracks in the TPC. For the HA-TPCs, the nominal value of 0.2 T is not sufficient. A three-dimensional magnetic-field map is required in the actual volume occupied by the chambers.

The leading direct effect of a field-scale error is a bias in the reconstructed momentum. To first order,

$$\frac{\delta p}{p} \simeq \frac{\delta B}{B}. \tag{2.5}$$

An incorrectly modelled field non-uniformity or field direction can additionally degrade the momentum resolution, introduce artificial dependencies on position and angle, and reduce the reliability of charge determination.

These considerations provide the scientific motivation for the development of a dedicated distributed magnetic-field monitoring system capable of collecting stable, spatially distributed and traceable magnetic-field measurements over long operating periods.

Chapter 3

System Requirements and Design Drivers

3.1 Overview

The ND280 Magnetic Field Monitoring System has been designed to satisfy a combination of scientific, functional, operational and engineering requirements.

The requirements defined in this chapter provide the reference framework for the hardware architecture, NodeOS firmware, ND280 Monitor application, validation activities and future system extensions.

Each requirement is assigned a unique identifier to support traceability throughout the development and qualification process.

3.2 Functional Requirements

ID	Requirement	Status
REQ-001	The system shall continuously acquire the three magnetic-field components B_x, B_y and B_z.	Verified
REQ-002	The system shall acquire the temperature measured by each magnetic sensor.	Verified
REQ-003	Each acquisition node shall provide a unique hardware identifier.	Verified
REQ-004	Calibration parameters shall be stored in non-volatile memory and shall survive power cycling.	Verified
REQ-005	Each acquisition node shall automatically recover from a firmware stall through hardware watchdog supervision.	Verified
REQ-006	Each acquisition node shall provide diagnostic information including acquisition statistics, communication errors and reset cause.	Verified
REQ-007	The monitoring software shall provide real-time visualization of magnetic-field and temperature measurements.	Verified
REQ-008	The monitoring software shall provide persistent data logging for long-duration acquisitions.	Verified
REQ-009	The command interface shall reject malformed or unrelated serial traffic without disrupting the acquisition process.	Verified
REQ-010	The magnetic-field measurement range shall provide adequate headroom with respect to the nominal ND280 magnetic field of approximately 0.2 T.	Verified

REQ-011	The local acquisition architecture shall support up to eight TMAG5273A2 sensors connected to a single intelligent node.	Rev.A design
REQ-012	The system architecture shall support multiple acquisition nodes connected through a common inter-board communication network.	In development
REQ-013	The distributed architecture shall support up to 32 acquisition boards, corresponding to a maximum of 256 magnetic sensors.	Design target
REQ-014	The same acquisition-board hardware shall be capable of operating either as a NODE or as a COORDINATOR.	Rev.A design
REQ-015	The NODE or COORDINATOR operating role shall be selected through a dedicated hardware configuration input.	Rev.A design

3.3 Reliability Requirements

ID	Requirement	Status
REL-001	The acquisition process shall continue without operator intervention during normal long-duration operation.	Verified
REL-002	A communication error shall not corrupt the internal measurement state of the acquisition node.	Verified
REL-003	Invalid EEPROM contents shall be detected and replaced by safe default calibration values.	Verified
REL-004	The firmware shall record the cause of the most recent microcontroller reset.	Verified
REL-005	The external I^2C interface of each board shall remain disabled during microcontroller reset and early boot.	Rev.A design
REL-006	The inter-board communication architecture shall electrically segment successive sections of the I^2C network.	Rev.A design

3.4 Engineering Design Drivers

The system architecture has also been influenced by a number of engineering design drivers that are not purely functional requirements but are considered essential for maintainability and future system evolution.

- **Modularity.** Hardware abstraction, device drivers, acquisition services and application logic shall remain separated.
- **Firmware reuse.** A new hardware revision should not require a complete redesign of NodeOS.
- **Deterministic embedded behaviour.** Dynamic memory allocation is avoided in the embedded firmware.
- **Hardware-defined identity.** The Node ID is determined by hardware configuration rather than by software or EEPROM contents.
- **Hardware-defined operating role.** The NODE/COORDINATOR role is determined by a dedicated hardware jumper.
- **Local data processing.** Each intelligent node performs sensor acquisition, averaging, calibration and diagnostics locally before data are transferred to higher-level controllers.

- **Independent subsystem validation.** Each hardware and software subsystem should be testable independently.
- **Incremental qualification.** New functionality is introduced only after the previous development stage has been validated.
- **Single-board architecture.** Whenever practical, NODE and COORDINATOR operation should use the same PCB and bill of materials.

3.5 Calibration and Metrological Requirements

The ND280 Magnetic Field Monitoring System is not intended to replace a dedicated high-precision magnetic-field mapping instrument. Its primary purpose is distributed, long-term monitoring of the magnetic field and the detection of spatial or temporal variations relevant to detector operation. Nevertheless, the usefulness of such a monitor depends on the measurements being sufficiently stable, reproducible and comparable between sensors and acquisition nodes.

A recent calibration study of a dedicated three-dimensional Hall sensor card developed at CERN illustrates the principal error sources that can affect Hall-based vector measurements, including offset, temperature dependence, non-linearity, cross-axis effects and mechanical or angular misalignment. The same study emphasizes that a dedicated calibration procedure is required when high measurement accuracy is needed [6]. The absolute performance target of that system is substantially more demanding than required for the ND280 monitoring application; however, the underlying metrological methodology is directly relevant to the present project.

The following requirements are therefore adopted for the calibration and characterization programme of the ND280 monitoring nodes:

ID	Requirement	Status
CAL-001	Each magnetic sensor shall have an individually identifiable calibration data set.	Design target
CAL-002	The calibration model shall support correction of per-axis offset and sensitivity and shall be extensible to cross-axis and alignment corrections if required by characterization data.	Design target
CAL-003	Calibration coefficients shall be stored in non-volatile memory together with a calibration format or version identifier.	Partially implemented
CAL-004	The calibration procedure shall characterize the temperature dependence of the magnetic-field measurement over the expected operating temperature range.	Planned
CAL-005	The calibration campaign shall evaluate sensor-to-sensor and board-to-board reproducibility using multiple independently assembled units.	Planned
CAL-006	The calibration model shall be validated over the magnetic-field range relevant to ND280 operation and shall provide adequate margin around the nominal field of approximately 0.2 T.	Planned
CAL-007	The system shall preserve the raw or minimally processed sensor information required to repeat calibration studies and to evaluate alternative correction models offline.	Design target
CAL-008	The calibration procedure and its resulting coefficients shall be traceable through documented calibration conditions, sensor identity, calibration date and validation results.	Planned

The calibration model will intentionally be kept no more complex than justified by the characterization results. The current offset-and-gain representation therefore remains a valid first-level model. A full three-dimensional correction matrix and explicit temperature terms will be introduced only if measurements demonstrate that they provide a significant improvement in the operating range of the monitoring system.

3.6 Scalability

The system has been intentionally developed through successive levels of complexity:

1. Reference Single-Sensor Node;
2. Eight-Sensor Multiplexing Node;
3. Distributed Multi-Board Acquisition System;
4. Complete detector monitoring infrastructure.

Each validated development stage provides the reference platform for the following stage.

The Reference Single-Sensor Node validates the complete acquisition chain and the NodeOS architecture.

The Eight-Sensor Multiplexing Node extends local acquisition capability by introducing the TCA9548A while preserving the existing firmware architecture.

The distributed system introduces inter-board communication and coordinator functionality without modifying the local sensor-acquisition concept.

3.7 Verification Strategy

Each requirement shall be associated with one or more verification activities.

Verification methods include:

- schematic and PCB design review;
- firmware unit and integration tests;
- communication protocol tests;
- watchdog and reset-recovery tests;
- EEPROM persistence tests;
- sensor acquisition tests;
- long-duration stability tests;
- inter-board communication tests;
- final detector-level characterization.

A Requirements Traceability Matrix will be maintained as part of the validation documentation and will associate each requirement with the corresponding implementation and verification evidence.

3.8 Summary

The requirements defined in this chapter establish the functional and engineering framework of the ND280 Magnetic Field Monitoring System.

The design prioritizes measurement continuity, robustness, modularity, scalability and long-term maintainability. These requirements directly motivate the system architecture described in the following chapter.

Chapter 4

System Architecture Views

4.1 Purpose of the Architecture Views

The architecture of the ND280 Magnetic Field Monitoring System is described through a set of complementary *System Views*. Each view answers one specific engineering question and intentionally omits implementation details that are not required at that level of abstraction.

This representation separates the overall system structure, the acquisition-node concept, the embedded-software organization, the measurement-processing path and the engineering evolution of the project. The same views can therefore remain valid across future hardware revisions even when individual devices or implementation details change.

4.2 Overall System Architecture

The complete monitoring infrastructure is organized as a hierarchy of intelligent acquisition nodes coordinated by a higher-level controller. The desktop application provides configuration, diagnostics, visualization and persistent data logging, while magnetic-field acquisition and preprocessing remain local to each node.

The distributed architecture deliberately avoids direct sensor access from the PC. Each acquisition node is responsible for local sensor management, calibration and diagnostics. This limits the amount of information exchanged over the inter-board network and isolates sensor-specific implementation details from the higher-level system.

4.3 Intelligent Acquisition Node

The Intelligent Acquisition Node is the fundamental building block of the distributed architecture. It combines a local sensor-acquisition path with an independent inter-board communication path, both managed by the same microcontroller and NodeOS firmware.

The separation of TWI1 and TWI0 is a central architectural choice. Sensor acquisition remains independent of the communication topology, allowing the local measurement subsystem to evolve without redefining the external node interface. The same PCB can operate as a NODE or as a COORDINATOR, with the role selected by a dedicated hardware input.

4.4 NodeOS Layered Architecture

NodeOS provides the common embedded-software architecture used by the acquisition nodes. The application layer is intentionally isolated from microcontroller registers and device-specific peripheral details.

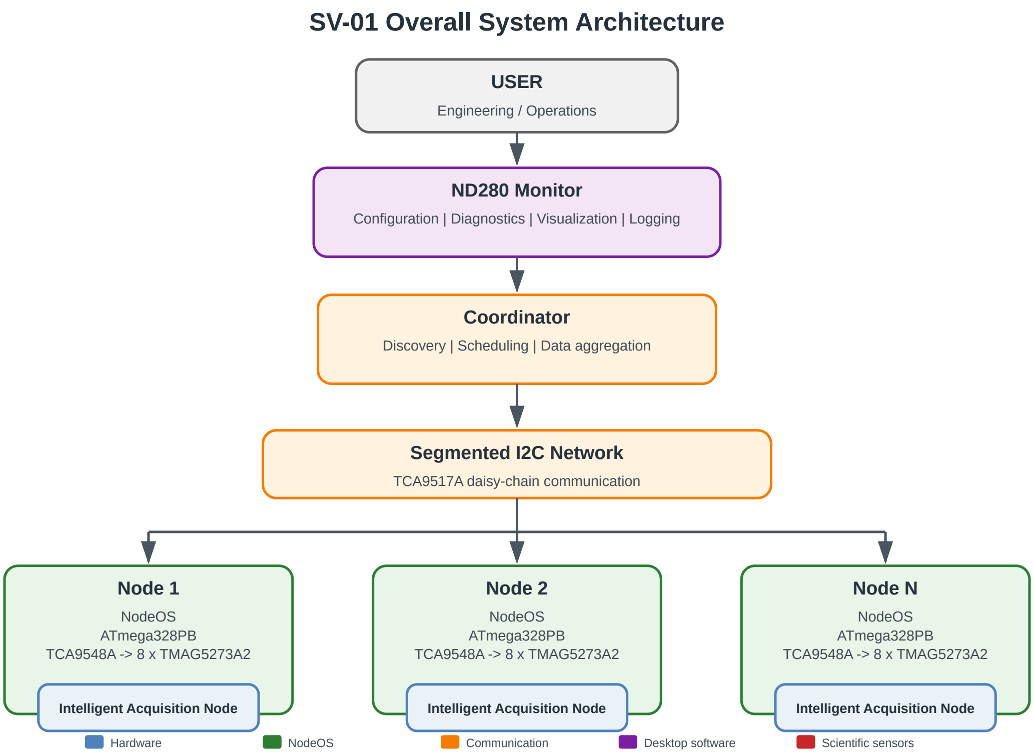

Figure 4.1: **SV-01 – Overall System Architecture.** The monitoring system is composed of autonomous intelligent acquisition nodes connected through a segmented communication network. A Coordinator performs node discovery and data aggregation, while the ND280 Monitor provides the user-facing engineering interface.

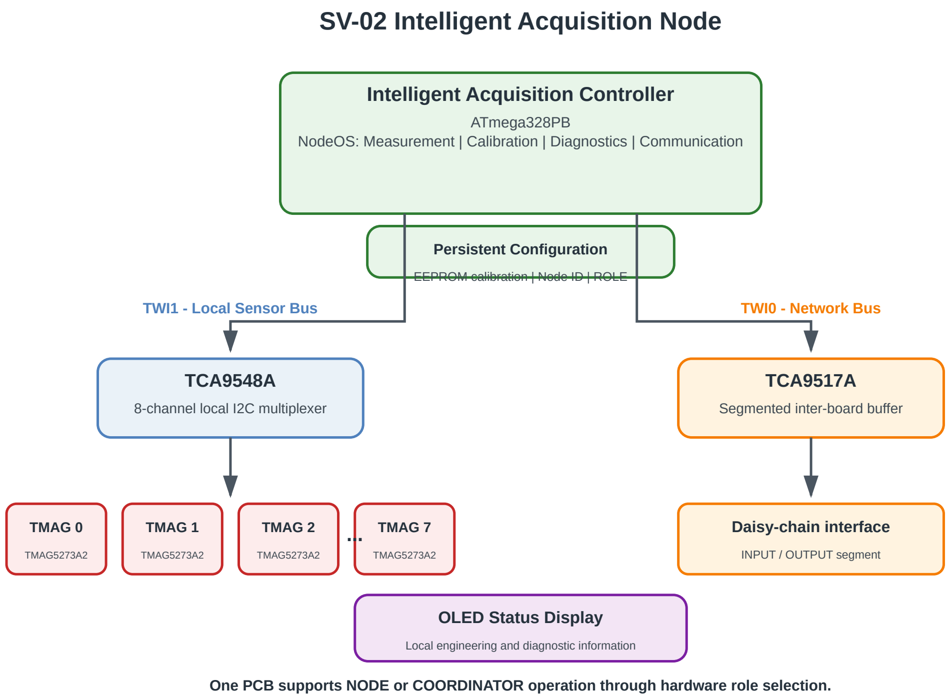

Figure 4.2: **SV-02 – Intelligent Acquisition Node.** TWI1 is dedicated to the local magnetic-sensor subsystem through the TCA9548A multiplexer, while TWI0 is reserved for the segmented inter-board network through the TCA9517A. Hardware identity, operating role, OLED status and persistent calibration are local node resources.

At the highest level, the application controls the acquisition sequence and system state. Core services implement measurement, calibration, diagnostics and communication. Platform services provide access to persistent storage, watchdog supervision, transport and peripheral drivers. The hardware-abstraction layer contains the microcontroller-dependent interface.

This separation is a design constraint rather than only a coding convention: the application layer does not directly access hardware resources.

4.5 Measurement Processing Workflow

The system converts a physical magnetic-field quantity into validated and traceable scientific information through a deterministic sequence of processing steps.

The TMAG5273A2 converts the local magnetic field and sensor temperature into digital samples. NodeOS performs acquisition, averaging and calibration, checks diagnostic status and constructs the measurement representation exposed to the communication layer. In the distributed system, the Coordinator aggregates measurements from multiple nodes before they are presented by the ND280 Monitor or stored as scientific data products.

This workflow preserves a clear distinction between raw sensor information, calibrated node-level measurements and higher-level aggregated data.

4.6 Engineering Evolution

The system has been developed incrementally. Each development stage was used to validate the assumptions required for the following stage rather than introducing all hardware and software complexity simultaneously.

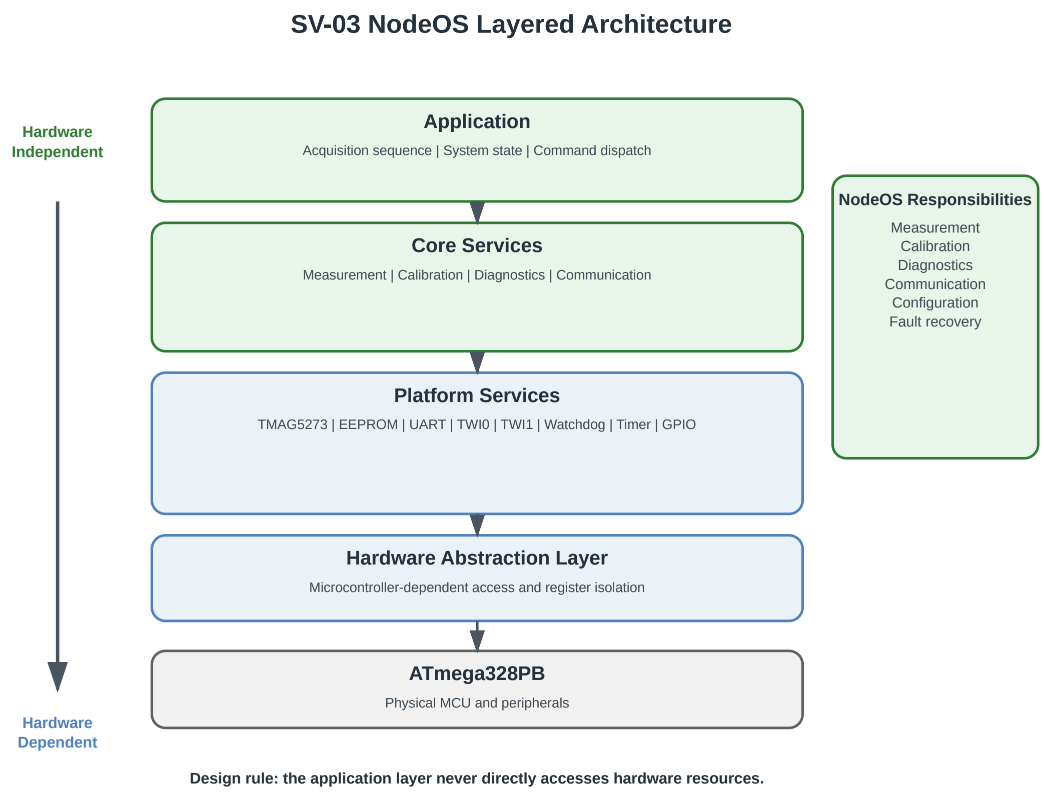

Figure 4.3: **SV-03 – NodeOS Layered Architecture.** Hardware abstraction and platform services isolate the application and core services from the underlying microcontroller resources. This organization allows hardware evolution with limited impact on high-level acquisition logic.

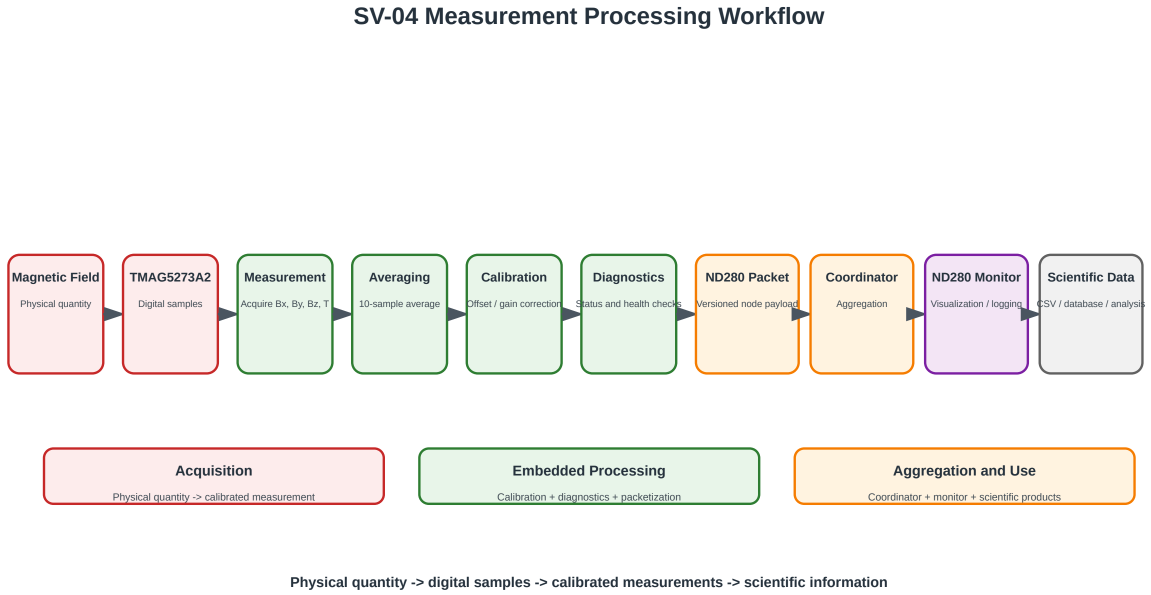

Figure 4.4: **SV-04 – Measurement Processing Workflow.** Transformation of the physical magnetic field into validated scientific information through acquisition, averaging, calibration, diagnostics, protocol framing, coordinator aggregation and desktop-level visualization or storage.

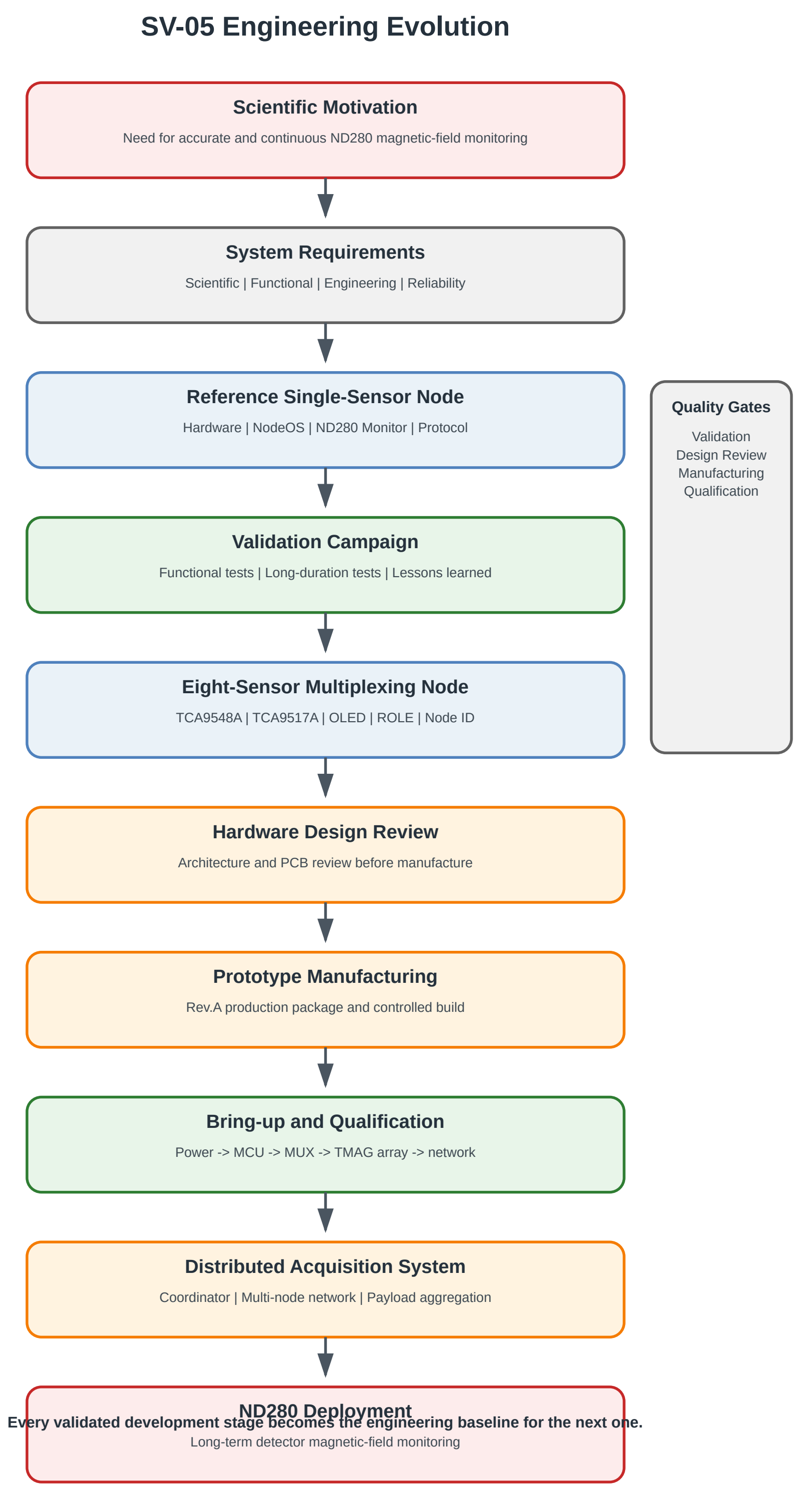

Figure 4.5: **SV-05 – Engineering Evolution.** Each validated development stage generates engineering knowledge and deliverables that become the baseline for the following system revision. The approach couples incremental development with validation, design review and qualification.

The Reference Single-Sensor Node first validated the complete acquisition chain. The resulting hardware, NodeOS firmware, communication protocol and ND280 Monitor then formed the reference platform for the Eight-Sensor Multiplexing Node. The Rev.A design adds local multiplexing and segmented inter-board communication without abandoning the validated architectural principles.

The next stage is prototype manufacturing and bring-up of the Eight-Sensor Multiplexing Node, followed by multi-board validation and Coordinator development. This process can be summarized by the following engineering principle:

> *Every validated development stage generates engineering knowledge that becomes the foundation of the following system revision.*

4.7 Summary

The System Views establish the common architectural vocabulary used throughout this report. The monitoring infrastructure is built from autonomous intelligent nodes; local sensor acquisition is separated from inter-board communication; NodeOS isolates high-level behaviour from hardware implementation; and measurement information is processed through a deterministic chain before aggregation and visualization.

The following chapter describes the Reference Single-Sensor Node, which was used to validate these principles before increasing hardware complexity.

Chapter 5

Reference Single-Sensor Node

The Reference Single-Sensor Node represents the engineering baseline of the ND280 Magnetic Field Monitoring System. Its objective was not to implement the final detector hardware, but to validate the complete acquisition chain before introducing multiplexing and distributed communication.

The platform was used to validate hardware operation, magnetic-field acquisition, persistent calibration, NodeOS, the serial communication protocol, the ND280 Monitor, watchdog recovery and long-duration operational behaviour. Every subsequent hardware revision is derived from the architecture validated on this reference platform.

5.1 Engineering Objectives

The first development stage was intentionally limited to one magnetic sensor. This reduced hardware complexity while preserving all functions required to exercise the complete system from the physical measurement to the desktop application.

The Reference Platform was required to validate:

- acquisition of the three magnetic-field components and sensor temperature;
- operation of the TMAG5273A2 sensor and the ATmega328PB TWI interface;
- the NodeOS layered firmware architecture;
- hardware node identification;
- persistent calibration in EEPROM;
- watchdog supervision and reset-cause diagnostics;
- deterministic serial command handling;
- the ND280 communication packet format;
- the ND280 Monitor desktop application;
- long-duration acquisition and logging.

The goal was therefore to establish a stable engineering reference, not merely to demonstrate that a magnetic sensor could be read.

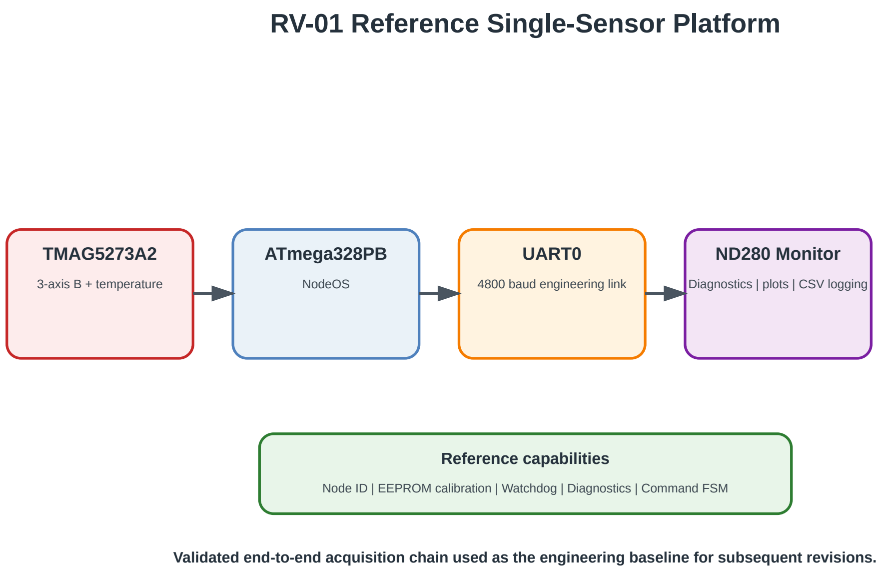

Figure 5.1: **RV-01 – Reference Single-Sensor Platform.** The reference acquisition chain couples one TMAG5273A2 sensor to the ATmega328PB running NodeOS and exposes the resulting measurements and diagnostics to the ND280 Monitor through UART.

5.2 Reference Platform Architecture

The hardware architecture was kept deliberately simple. The ATmega328PB provides the embedded processing and peripheral interfaces, while the TMAG5273A2 provides three-axis magnetic-field and temperature measurements. The sensor is operated in the A2 range configuration used by the project, corresponding to a full-scale magnetic range of approximately 266 mT.

The reference firmware configuration uses ten samples per reported measurement and a 100 ms interval between samples. The measured values are averaged before calibration and packet generation. During the validation campaign the node communicated with the engineering PC interface through UART0 at 4800 baud.

The board includes hardware Node ID selection, non-volatile calibration storage and watchdog supervision. These functions were introduced at the reference stage because they were considered architectural requirements for the later distributed system rather than optional debugging features.

5.3 NodeOS Integration

The Reference Platform was the first complete hardware implementation of NodeOS. The firmware architecture separates application behaviour, reusable services, platform services and hardware-dependent access.

On the reference node, NodeOS provides:

- TMAG5273A2 initialization and measurement acquisition;
- sample averaging;
- calibration offset and gain application;
- EEPROM persistence and calibration-state reporting;

- hardware Node ID management;
- UART packet generation and command reception;
- diagnostics and internal counters;
- watchdog servicing and reset-cause reporting.

Developing and stabilizing these services before introducing the eight-channel multiplexer substantially reduced the risk of mixing firmware-architecture problems with new hardware-integration problems.

5.4 Communication Architecture

The reference node uses UART0 as its engineering communication interface. Measurement data are transmitted using the `$ND280` packet family, while diagnostic and calibration information use dedicated response families such as `$ND280DIAG`, `$ND280CAL` and `$ND280RSP`.

The command receiver evolved during validation. Early line-based parsing could interpret serial noise, echoes or unrelated traffic as commands. This behaviour was addressed by introducing a deterministic finite-state-machine parser in NodeOS v0.5.3-alpha1.

Commands are explicitly framed by the colon character. Examples include:

```
:HELP
:INFO
:DIAG
:CAL READ
:DIAG WDT TEST
```

Input that does not belong to a valid command frame is ignored. The same rule is implemented by the ND280 Monitor transport layer, which adds the command prefix before transmission.

5.5 Validation Campaign

The Reference Platform was subjected to repeated functional tests and extended unattended acquisition runs. Validation covered the sensor interface, calibration persistence, watchdog recovery, serial communication, command parsing, diagnostics and desktop logging.

A representative long-duration run reached an uptime of 68,395 seconds (approximately 19 hours). At the end of the run the diagnostic counters reported 49,425 successful measurements, zero measurement failures, zero TWI errors and zero EEPROM errors.

Table 5.1: Representative Reference Node long-duration validation results.

Parameter	Observed value
Uptime	68,395 s
Successful measurements	49,425
Measurement failures	0
TWI errors	0
EEPROM errors	0

The run demonstrated stable sensor acquisition and persistence behaviour. UART receive robustness was treated as a separate engineering issue: receive-overflow and parser behaviour observed during earlier tests led to UART hardening and ultimately to the finite-state-machine command parser. The corrected firmware was subsequently exercised with diagnostic commands without disturbing normal measurement acquisition.

Watchdog functionality was also verified explicitly using the diagnostic watchdog test command. The firmware acknowledged the test, intentionally stopped servicing the watchdog and restarted with the reset cause correctly reported as `WATCHDOG`. This verified both the recovery mechanism and the associated diagnostics.

5.6 Engineering Lessons Learned

The Reference Platform produced several design decisions that directly influenced the later Eight-Sensor Multiplexing Node.

Validate the complete acquisition chain before increasing hardware complexity

The single-sensor platform allowed sensor, firmware, protocol and desktop-software problems to be isolated and corrected without the additional variables introduced by multiplexing or multi-board networking.

Firmware architecture must precede hardware evolution

NodeOS was stabilized before the eight-sensor hardware was introduced. The new board can therefore extend the measurement service instead of requiring a complete firmware redesign.

Diagnostics are part of the system architecture

Watchdog counters, reset flags, TWI statistics, measurement counters and EEPROM status proved essential during long-duration testing. They are retained as permanent services rather than temporary debug instrumentation.

Command processing must be deterministic

Serial command reception must be explicitly framed and independent of arbitrary serial traffic. The finite-state-machine parser and mandatory colon prefix were introduced as a direct consequence of validation observations.

Calibration and identity must survive operational cycles

Persistent EEPROM calibration and hardware-defined Node ID prevent configuration ambiguity after power cycling and simplify replacement or reinstallation of a node.

Long-duration tests are required before freezing the next hardware stage

Short functional tests demonstrated correctness, but unattended operation exposed communication and recovery behaviours that were not visible during bring-up. Long-duration validation therefore became a required gate before subsequent hardware evolution.

5.7 Summary

The Reference Single-Sensor Node established the validated baseline of the ND280 Magnetic Field Monitoring System. It demonstrated that the sensor, NodeOS firmware, communication protocol, calibration storage, watchdog diagnostics and ND280 Monitor could operate as one coherent acquisition chain over extended periods.

The resulting engineering knowledge and validated software architecture formed the basis of the Eight-Sensor Multiplexing Node described later in this report.

Chapter 6

NodeOS Firmware

NodeOS is the embedded firmware architecture developed for the acquisition nodes. It separates hardware abstraction, device drivers, services and application logic.

The current validated baseline includes magnetic-field and temperature acquisition, averaging, EEPROM calibration storage, diagnostics, watchdog recovery, reset-cause reporting and a deterministic finite-state-machine command parser.

Chapter 7

ND280 Monitor

The ND280 Monitor is the Python desktop application used for commissioning, diagnostics, calibration, real-time visualization and CSV logging.

The monitor implements the command transport required by the NodeOS finite-state-machine parser and provides engineering access to node identity, diagnostics and calibration information.

7.1 Operational View

The application provides a unified view of the selected node, the magnetic-field and temperature trends, the NodeOS control panel, diagnostic responses and the raw serial stream. This makes it possible to correlate high-level monitoring information with the exact protocol traffic produced by the embedded node.

Figure 7.1 shows the ND280 Monitor during an extended acquisition run of the Reference Single-Sensor Node. The screenshot documents simultaneous visualization of the three magnetic-field components, sensor temperature, node identity, packet continuity and the NodeOS diagnostic response.

7.2 Engineering Role

The ND280 Monitor is not only a visualization tool. During development it has been used as the primary engineering interface for firmware bring-up, command testing, EEPROM and calibration operations, long-duration logging and diagnostic inspection. Its behaviour therefore forms part of the validated end-to-end acquisition chain described in this report.

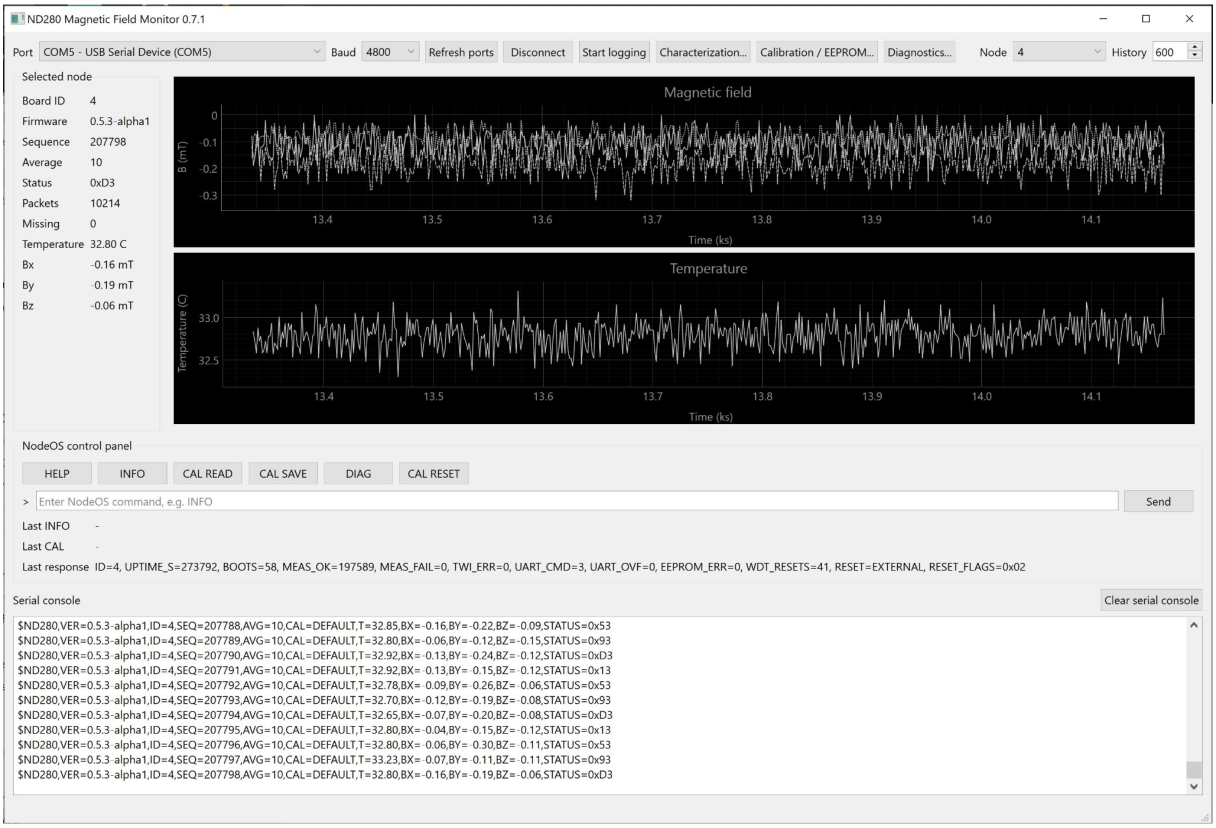

Figure 7.1: ND280 Monitor during an extended long-duration acquisition test of the Reference Single-Sensor Node. At the time of the screenshot, Node 4 running NodeOS v0.5.3-alpha1 reported an uptime of 273792 s, 197589 successful measurements, no measurement failures, no TWI errors, no UART receive overflow and no EEPROM errors. The monitor simultaneously displays magnetic-field and temperature trends, node status, the diagnostic response and the raw NodeOS serial stream.

Chapter 8

Eight-Sensor Multiplexing Node

The Eight-Sensor Multiplexing Node extends the Reference Node by adding a TCA9548A local I^2C multiplexer and up to eight TMAG5273A2 sensors.

The Rev.A hardware also includes a TCA9517A for segmented inter-board communication, an OLED display, a hardware NODE/COORDINATOR role selector, hardware node identification and dedicated reset/enable control signals.

8.1 Rev.A Hardware Realization

Figure 8.1 shows the current three-dimensional rendering of the Eight-Sensor Multiplexing Node Rev.A. The rendering is included to connect the functional architecture introduced in Chapter 4 with the physical hardware implementation. The eight TMAG sensor positions are distributed along the lower edge of the board, while the controller, local multiplexer, OLED display, configuration elements and inter-board connectors are integrated on the same PCB.

8.2 Architectural Continuity

The Rev.A board preserves the architectural principles validated on the Reference Single-Sensor Node. Local acquisition remains under NodeOS control, while the additional hardware extends the number of sensors and introduces the physical infrastructure required for distributed operation. This allows the new board to be treated as an evolution of the validated reference platform rather than as an independent redesign.

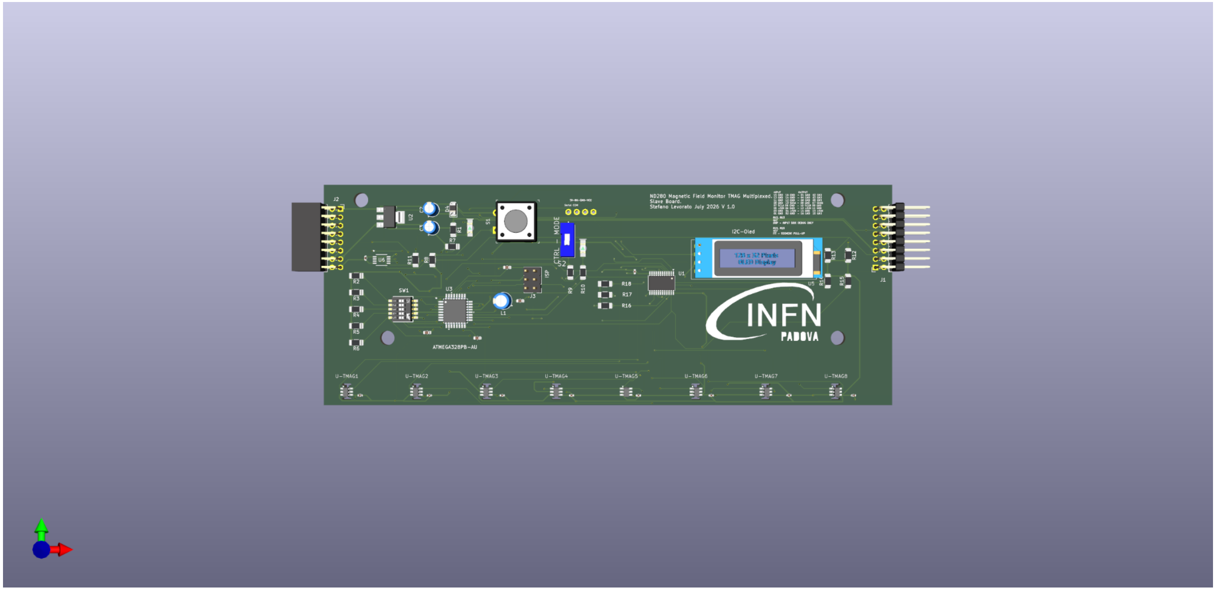

Figure 8.1: Three-dimensional rendering of the Eight-Sensor Multiplexing Node Rev.A. The board integrates the ATmega328PB controller, TCA9548A local sensor multiplexer, TCA9517A segmented inter-board interface, OLED display, hardware configuration elements and eight TMAG5273A2 sensor positions on a single PCB.

Chapter 9

Distributed Acquisition System

The distributed architecture is intended to support multiple acquisition boards connected in a segmented daisy chain.

Each NODE performs local acquisition and exposes processed measurements to the upstream COORDINATOR. The COORDINATOR scans the configured node-address range, maintains a node table and aggregates measurements into a coherent higher-level payload.

The long-term target is a system of up to 32 acquisition boards, corresponding to a maximum of 256 magnetic sensors.

Chapter 10

Communication Protocol

The current NodeOS command protocol uses a mandatory colon prefix. Measurement and diagnostic responses use the `$ND280` packet family.

Examples include:

```
:HELP
:INFO
:DIAG
:CAL READ
```

The protocol will be extended for multi-sensor and multi-board operation while preserving explicit versioning.

Chapter 11

Validation Campaign

The Reference Node has been exercised through functional tests and long-duration acquisition runs.

11.1 Initial Long-Duration Test

A representative long-duration test reached an uptime of 68,395 seconds, with 49,425 successful measurements, zero measurement failures, zero TWI errors and zero EEPROM errors.

UART receive robustness was subsequently improved through the finite-state-machine command parser used in NodeOS v0.5.3-alpha1.

11.2 Extended Long-Duration Test

A later extended acquisition run provided a substantially longer operational sample. As documented in Figure 7.1, the Reference Single-Sensor Node reached an uptime of 273792 seconds (approximately 76 hours). At the captured point the diagnostic counters reported 197589 successful measurements, zero measurement failures, zero TWI errors, zero UART receive overflows and zero EEPROM errors.

The ND280 Monitor simultaneously reported zero missing packets for the displayed acquisition sequence. This extended run provides additional evidence that the acquisition, TWI communication, EEPROM handling, command transport and monitoring chain can operate continuously over multi-day periods without the error modes observed during the earlier UART parser development phase.

These results validate the Reference Node as an engineering baseline. Final noise, offset, thermal behaviour, absolute calibration and distributed-bus performance remain to be qualified on the Eight-Sensor Multiplexing Node and on the final detector hardware.

11.3 Preliminary Stability Characterization

Before a controlled metrological calibration campaign is performed, data acquired with the Reference Single-Sensor Node already provide a useful first characterization of the observed short-term stability of the complete acquisition chain. The results presented in this section are based on a continuous dataset containing 34,418 magnetic-field measurements. They should be interpreted as *preliminary system-level observations*: they include the sensor, embedded averaging and readout chain, local environmental conditions and any real variation of the ambient magnetic field. They are therefore not measurements of the intrinsic TMAG5273A2 resolution or of its absolute accuracy.

11.3.1 Observed field distributions

Figure 11.1 shows the distributions of the three measured field components. After rebinning, Gaussian functions were fitted as a compact first description of the observed widths. The fitted standard deviations are approximately 0.0427 mT, 0.0464 mT and 0.0314 mT for B_x, B_y and B_z, respectively. The corresponding fitted means are approximately -0.0944 mT, -0.2921 mT and 0.1228 mT.

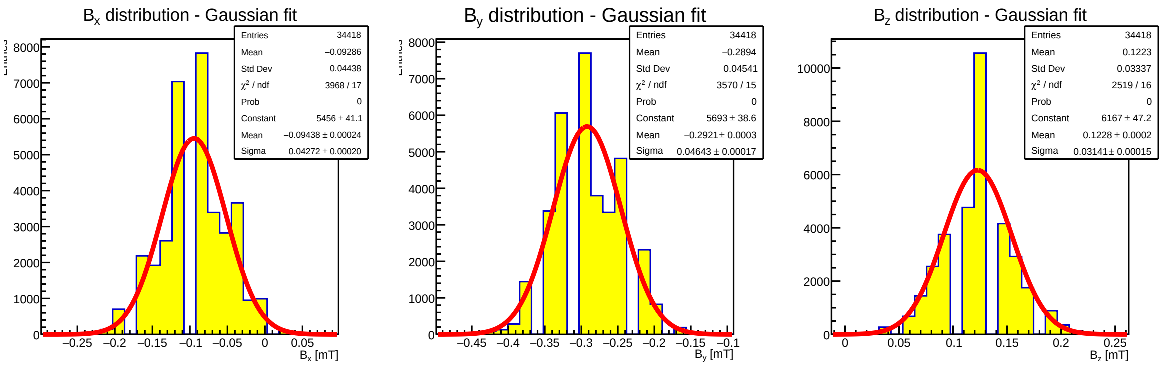

Figure 11.1: Preliminary distributions of the three magnetic-field components measured with the Reference Single-Sensor Node over 34,418 samples. Gaussian fits are superimposed after rebinning and are used only as a first characterization of the observed distribution widths.

The Gaussian description is not statistically adequate for the complete distributions: the fitted χ^2/ndf values are approximately 3968/17, 3570/15 and 2519/16 for B_x, B_y and B_z, respectively. Consequently, the fitted σ values must not be interpreted as intrinsic sensor resolutions. The non-Gaussian structure may contain contributions from quantization, slow drift, temperature dependence, environmental magnetic-field variations and other effects of the complete measurement chain. This observation directly motivates the controlled characterization programme described later in this chapter.

11.3.2 Temporal behaviour

The same dataset was inspected as a function of acquisition time. Figure 11.2 overlays the three measured field components throughout the run. No correction for temperature or slow environmental variation has been applied in this representation.

This view is complementary to the one-dimensional distributions because it allows abrupt excursions, slow drifts and time-correlated structures to be distinguished from purely point-to-point fluctuations. A quantitative stability specification will only be assigned after equivalent measurements have been repeated under controlled magnetic-field and temperature conditions.

11.3.3 Preliminary temperature dependence

The recorded TMAG5273A2 temperature allows a first investigation of whether changes in the reported magnetic-field components are correlated with sensor temperature. Figure 11.3 shows profiles of the mean B_x, B_y and B_z values versus temperature for the same dataset. The sampled temperature interval is approximately 26 °C to 32 °C, with a mean close to 28 °C.

Any apparent slope in these profiles must be interpreted cautiously. In an uncontrolled run, temperature, elapsed time and the ambient magnetic field are not independent variables. A correlation between B and temperature can therefore arise from sensor response, from a real environmental field change correlated with time, or from both. The purpose of this preliminary analysis is to identify effects that must subsequently be separated in a controlled calibration campaign, not to derive final temperature-correction coefficients.

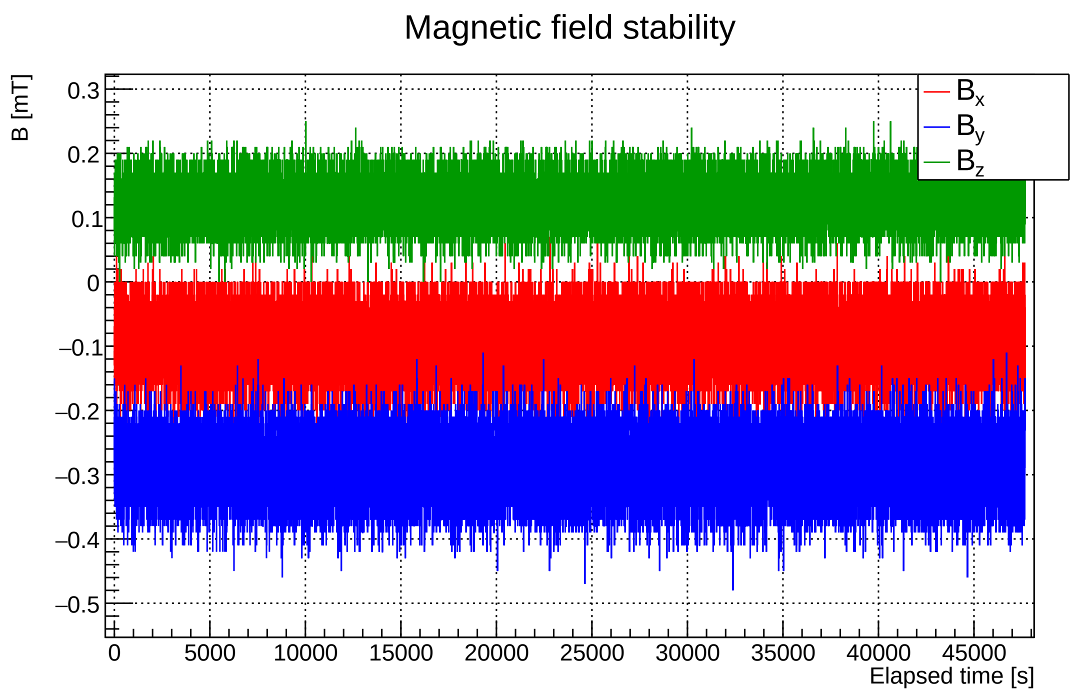

Figure 11.2: Preliminary time evolution of B_x, B_y and B_z during the stability dataset. The plot represents the measured values before a dedicated temperature-dependent or metrological correction is applied.

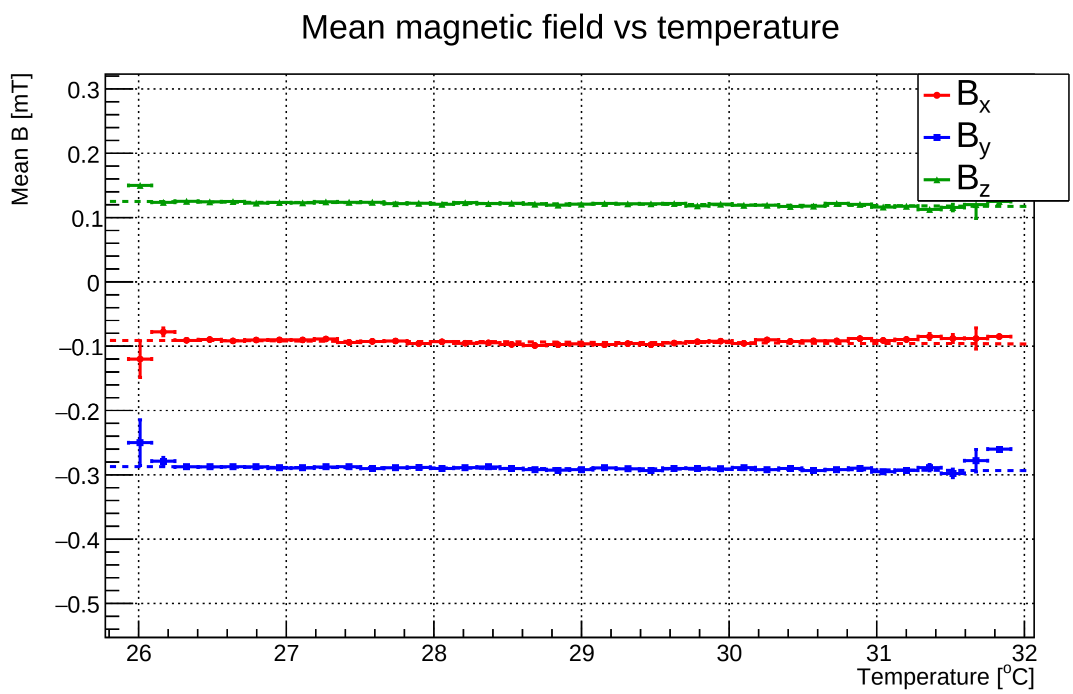

Figure 11.3: Profile of the mean measured magnetic-field components versus TMAG5273A2 temperature. The profiles are intended to reveal possible temperature-correlated behaviour in the complete acquisition chain; they do not by themselves determine the intrinsic temperature coefficients of the sensor.

Taken together, the distributions, time evolution and temperature profiles establish a useful pre-calibration baseline for the Reference Node. They also demonstrate why long-duration stability and metrological characterization are complementary: operational continuity alone does not establish measurement accuracy, while a calibration model must be supported by evidence that it remains stable during realistic operation.

11.4 Calibration and Metrological Characterization Strategy

11.4.1 Scope of the calibration programme

The ND280 Magnetic Field Monitoring System is intended primarily as a distributed monitoring instrument rather than as an absolute magnetic-field mapping system. Its calibration strategy must therefore be proportional to this objective. The system does not need to reproduce the metrological performance of dedicated high-precision Hall mapping probes; however, calibration cannot be neglected because sensor-to-sensor differences, offsets, temperature dependence and cross-axis effects could otherwise generate apparent field variations that are unrelated to the detector magnet.

A useful reference is the calibration and characterization procedure developed for a dedicated CERN 3D Hall sensor card [6]. That work identifies offset, temperature dependence, non-linearity, planar Hall effects and mechanical misalignment as relevant contributors to the final measurement error, and demonstrates the value of a controlled calibration procedure for compensating such effects. The authors report mean residuals of approximately 9.15×10^{-5} T in field magnitude and 0.97 mrad in angle over a 0.1–2 T range across ten calibrated units. These values define the performance of a substantially more demanding mapping application and are not adopted as requirements for the ND280 monitor; they nevertheless provide a useful example of how calibration accuracy, temperature stability and production reproducibility can be quantified.

For the ND280 system, the principal metrological objectives are instead:

- repeatable measurement of temporal changes in the magnetic field;
- comparability of measurements from different TMAG5273A2 sensors and acquisition boards;
- controlled dependence of the reported field on sensor temperature;
- sufficient absolute accuracy to identify incorrect installation, configuration or gross field deviations;
- traceable calibration parameters that remain associated with the physical sensor throughout operation and maintenance.

11.4.2 Proposed calibration model

The present NodeOS implementation supports per-axis offset and gain parameters stored in non-volatile memory. This representation is intentionally simple and remains appropriate for early bring-up and engineering tests. The calibration infrastructure will, however, be designed so that it can evolve without changing the higher-level acquisition protocol.

The natural next step is a vector calibration model of the form

$$\vec{B}_{\text{cal}} = \mathbf{M}\left(\vec{B}_{\text{raw}} - \vec{O}\right), \tag{11.1}$$

where $\vec{O}$ is a three-component offset vector and $\mathbf{M}$ is a 3×3 correction matrix. The diagonal terms of $\mathbf{M}$ describe axis sensitivities, while the off-diagonal terms can absorb first-order cross-axis coupling and small orientation errors. A temperature-dependent extension may be introduced after characterization, for example by allowing the offset vector and selected matrix coefficients to vary around a reference temperature T_0.

The complexity of the final model will be determined experimentally. The CERN Hall-card study uses a considerably more sophisticated response model based on spherical harmonics because it addresses discrete Hall elements, mechanical alignment and a very wide field range [6]. Such a model would be unnecessarily complex for the present system unless the TMAG5273A2 characterization demonstrates residual non-linearities that cannot be corrected with a simpler matrix-based approach.

11.4.3 Field-range characterization

The dedicated Hall-card study shows that calibration parameters may vary with magnetic-field strength and describes a multi-field procedure in which calibration coefficients determined at different field magnitudes are interpolated across the operating range [6]. This observation is relevant even though the ND280 monitoring system operates in a much narrower field region centered near 0.2 T.

The first Rev.A characterization campaign should therefore include several reference field values around the expected operating region rather than relying on a single-point calibration. A preliminary set of characterization points may include zero field and representative values around 0.1 T, 0.2 T and 0.25 T, subject to the capabilities of the available reference magnet and instrumentation. These values are not yet calibration requirements; they define an initial measurement programme intended to determine whether a single linear correction is sufficient in the ND280 operating range.

If the residuals show a significant field-dependent structure, the calibration model can be extended either by piecewise interpolation of coefficients or by an alternative non-linear correction. If no significant improvement is observed, the simpler model will be retained.

11.4.4 Temperature characterization

Temperature must be treated as a measured calibration variable rather than only as an environmental diagnostic. This requirement is facilitated by the TMAG5273A2, which already provides local temperature information for every sensor.

The CERN calibration study demonstrates why this is important: the residual error in reconstructed field magnitude changes measurably between the calibration temperatures and the temperature extremes, even after application of a temperature correction model [6]. The ND280 characterization campaign should therefore repeat representative field measurements at several temperatures spanning the expected detector operating range. The objective is not to reproduce the temperature-controlled apparatus of the reference study, but to quantify the temperature coefficient and determine whether explicit compensation is required for the desired monitoring precision.

11.4.5 Sensor-to-sensor and board-to-board reproducibility

A distributed system containing potentially hundreds of sensors cannot rely on the behaviour of a single characterized device. The calibration campaign must therefore quantify the spread between TMAG5273A2 devices and between independently assembled acquisition boards.

The reference Hall-card work calibrated and tested ten separate cards and used the dispersion of the reconstructed field residuals as a measure of production and calibration reproducibility [6]. A similar principle should be adopted for the ND280 system. After Rev.A bring-up, multiple boards should be exposed to identical magnetic-field and temperature conditions and their residual distributions compared. This will determine whether one generic calibration model is sufficient, whether coefficients must be determined sensor-by-sensor, and how much of the measurement uncertainty originates from production variability.

11.4.6 Calibration data management

Calibration data shall remain permanently associated with the physical sensor. The current EEPROM infrastructure already provides the basis for persistent storage, but the data format should evolve from the present simple gain-and-offset representation to a versioned calibration record. A future calibration record should contain, as appropriate:

- calibration-model version;
- sensor and board identifiers;
- three-axis offset vector;
- three-dimensional correction matrix;
- temperature coefficients and reference temperature;
- calibration date and calibration-range identifier;
- validation or quality flags;
- integrity information such as a CRC.

Versioning is important because it allows NodeOS to retain backward compatibility with the existing Reference Node while supporting more complete calibration models on later hardware revisions.

11.4.7 In-situ verification

Laboratory calibration and detector operation serve different purposes. Laboratory calibration determines the relationship between sensor output and a known magnetic field under controlled conditions. Once installed in ND280, the monitoring network should instead perform periodic consistency checks intended to reveal drift, installation problems or anomalous sensors.

In-situ checks may include comparisons between nearby sensors, repeated measurements at reproducible magnet operating points, verification of zero-field offsets when such operating conditions are available, and comparison with the established ND280 magnetic-field map. These procedures do not replace laboratory calibration; they provide long-term verification that the calibration remains applicable throughout detector operation.

11.4.8 Calibration work package for Rev.A

The calibration programme will therefore be introduced as a dedicated work package following the electrical bring-up of the Eight-Sensor Multiplexing Node. The initial objective is characterization rather than immediate implementation of a complex correction algorithm. Raw data from multiple sensors, field values and temperatures will first be collected and analysed offline. Only after the dominant error terms have been identified will the corresponding correction model be frozen and incorporated into NodeOS and the EEPROM calibration format.

This staged approach is consistent with the overall engineering philosophy of the project: measurement evidence determines the complexity of the implementation, and each additional calibration feature must demonstrate a measurable improvement before becoming part of the reference architecture.

Chapter 12

Conclusions and Future Developments

The Reference Single-Sensor Node has validated the core hardware, firmware and software architecture.

The next project phase is the prototype manufacture and staged bring-up of the Eight-Sensor Multiplexing Node Rev.A, followed by progressive validation of the inter-board communication architecture and the coordinator functionality.

Following electrical and firmware bring-up, a dedicated calibration and metrological characterization work package will quantify sensor offsets, relative sensitivity, cross-axis behaviour, temperature dependence, reproducibility and long-term stability in the magnetic-field range relevant to ND280. The calibration model will be selected from measurement evidence and will remain deliberately simpler than the procedures required for high-precision field-mapping instruments unless the characterization data demonstrate the need for additional correction terms.

The resulting programme will provide a traceable basis for interpreting long-term changes measured by the distributed monitoring network and for comparing measurements from different sensors and acquisition nodes.

Appendix A

Hardware Revision History

Revision	Date	Status	Description
Reference Node	2026	Validated	Single-sensor reference platform
Rev.A	2026	Prototype design	Eight-sensor multiplexing board

Appendix B

Firmware Revision History

Version	Status	Main content
0.5.2-alpha1a	Stable baseline	Watchdog and UART RX hardening
0.5.3-alpha1	Stable baseline	Finite-state-machine command parser

Appendix C

Protocol Reference

C.1 Command Prefix

All NodeOS commands shall begin with the ASCII character ':'.

C.2 Current Commands

```
:HELP
:INFO
:DIAG
:DIAG WDT TEST
:CAL READ
:CAL OFFSET x y z
:CAL GAIN x y z
:CAL NOISE x y z
:CAL TEMP t
:CAL SAVE
:CAL RESET
```

Appendix D

Architecture Decision Record Index

The Architecture Decision Records are maintained separately in the project repository. The TDR references the accepted decisions relevant to each subsystem.

Bibliography

[1] K. Aivazelis and others. Performance of the High-Angle Time Projection Chambers in the Upgraded T2K Off-Axis Near Detector. T2K Collaboration, preprint, 2025.

[2] T2K Collaboration. A dedicated device for measuring the magnetic field of the ND280 magnet in the T2K experiment. *Journal of Instrumentation*. Publication information available from CERN Document Server.

[3] T2K Collaboration. The TPCs of the T2K experiment. Workshop presentation.

[4] T2K Collaboration. New TPCs for T2K near detector. Presentation.

[5] T2K Collaboration. Document for ND280 review. Technical report.

[6] L. Messner, N. Pacifico, D. Baumgarten, and R. Dumps. Calibration and characterization of a 3D magnetic Hall sensor card. Prepared for submission to JINST; CERN EP-DT and University of Innsbruck.